Formula Sheet

Random Walks & Gambler's Ruin

Chapter 4.3 · Jane Street National Probability Challenge · Stage 1

0. Notation (read once)

Symbols used everywhere

- **Walk:** $S_n = X_1 + \cdots + X_n$, $X_i = +1$ w.p. p and -1 w.p. $q = 1-p$, $S_0 = 0$. Fair walk: $p = q = \frac{1}{2}$.
- **Drift and variance per step:** $\mu = p - q$, $\sigma^2 = \mathrm{Var}(X) = 1 - \mu^2 = 4pq$.
- **Odds ratio:** $r = q/p$. $M_n = \max_{0 \le i \le n} S_i$. $T_a = \min\{n : S_n = a\}$ (first hitting time).
- **Gambler's ruin:** fortune $k \in \{0, \ldots, N\}$, absorbing at 0 (ruin) and N (goal). $W_k = \mathbb{P}(\text{reach } N \text{ before } 0)$, $Q_k = 1 - W_k$, $D_k = \mathbb{E}[\text{duration}]$, $T =$ exit time.
- **Counting:** $\#\text{paths}(n \to j) = \binom{n}{(n+j)/2}$ (needs $n + j$ even, $|j| \le n$). $u_{2n} = \binom{2n}{n} 4^{-n}$. $C_n = \frac{1}{n+1}\binom{2n}{n}$.

1. Simple random walk: distribution

1.1 Where is the walk after n steps?

$$\mathbb{P}(S_n = j) = \binom{n}{\frac{n+j}{2}} p^{\frac{n+j}{2}} q^{\frac{n-j}{2}}, \qquad \text{fair: } \binom{n}{\frac{n+j}{2}} 2^{-n}.$$

$$\mathbb{E}[S_n] = n\mu = n(p-q), \qquad \mathrm{Var}(S_n) = 4pq\, n \quad (= n \text{ if fair}).$$

Parity rule: $S_n \equiv n \pmod 2$. Odd levels are impossible at even times, so "$S_{20} \ge 5$" means "$S_{20} \ge 6$".

Normal approximation (fair): $S_n \approx \mathcal{N}(0, n)$, and the local version $\mathbb{P}(S_n = j) \approx \dfrac{2}{\sqrt{2\pi n}} e^{-j^2/2n}$ (factor 2 because of parity). Also $\mathbb{E}|S_n| \approx \sqrt{2n/\pi}$.

2. Return to the origin (syllabus: "return to origin")

2.1 Fair walk: return probabilities

$$u_{2n} = \mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 4^{-n} \;\approx\; \frac{1}{\sqrt{\pi n}}, \qquad u_{2n} = u_{2n-2} \cdot \frac{2n-1}{2n}.$$

$$f_{2n} := \mathbb{P}(\text{first return at time } 2n) = u_{2n-2} - u_{2n} = \frac{u_{2n}}{2n-1} = \frac{2C_{n-1}}{4^n}.$$

$$\mathbb{P}(S_k \ne 0 \ \forall\, 1 \le k \le 2n) = \mathbb{P}(S_k \ge 0 \ \forall\, k \le 2n) = u_{2n}, \qquad \mathbb{P}(S_k > 0 \ \forall\, 1 \le k \le 2n) = \tfrac{1}{2} u_{2n}.$$

Generating functions: $\sum_n u_{2n} x^n = (1-x)^{-1/2}$ and $\sum_n f_{2n} x^n = 1 - \sqrt{1-x}$. Hence $\sum_n f_{2n} = 1$ (return is certain) but $\mathbb{E}[\text{return time}] = \infty$.

Expected visits to 0 in the first $2n$ steps (counting time 0): $\sum_{k=0}^{n} u_{2k} = (2n+1)\, u_{2n} \approx 2\sqrt{n/\pi}$.

2.2 Biased walk: return probabilities

$$\mathbb{P}(\text{ever return to 0}) = 1 - |p-q| = 2\min(p,q), \qquad \mathbb{E}[\#\text{visits to 0, incl. start}] = \sum_n u_{2n} = \frac{1}{|p-q|}.$$

(Number of visits is geometric: each visit is followed by another with probability $2\min(p,q)$.)

3. The reflection principle

3.1 Core statement and its consequences (fair walk, level $m > 0$)

Reflection: the number of n-step paths from 0 to $e < m$ that touch m equals the number of paths from 0 to $2m - e$.

$$\mathbb{P}(M_n \geq m,\ S_n = e) = \begin{cases} \mathbb{P}(S_n = 2m - e), & e \leq m, \\ \mathbb{P}(S_n = e), & e \geq m. \end{cases}$$

$$\boxed{\mathbb{P}(M_n \geq m) = \mathbb{P}(S_n \geq m) + \mathbb{P}(S_n > m) = 2\mathbb{P}(S_n > m) + \mathbb{P}(S_n = m)}$$

$$\mathbb{P}(M_n = m) = \mathbb{P}(S_n = m) + \mathbb{P}(S_n = m+1) \quad \text{(only one term is nonzero, by parity).}$$

Hitting-time distribution: $\mathbb{P}(T_a \leq n) = \mathbb{P}(M_n \geq a)$, and the *hitting-time theorem*

$$\mathbb{P}(T_a = n) = \frac{a}{n}\,\mathbb{P}(S_n = a) = \frac{a}{n}\binom{n}{\frac{n+a}{2}} 2^{-n} \qquad (a \geq 1).$$

By symmetry the same formulas hold for the minimum with $m \to -m$.

3.2 Advanced: two barriers (alternating reflections)

The number of n-step paths from 0 to e that stay strictly inside $(-a, b)$ is

$$\sum_{k\in\mathbb{Z}} \Big[\, \#(n \to e + 2k(a+b)) \;-\; \#(n \to 2b - e + 2k(a+b)) \Big],$$

where $\#(n \to j) = \binom{n}{(n+j)/2}$ (and 0 if impossible). Reflect first across one wall, then across the other, and so on, correcting by inclusion–exclusion.

4. Ballot theorem and Catalan numbers

4.1 Ballot problem (a votes for A, b for B, $a > b$, random order)

$$\mathbb{P}(\text{A strictly ahead throughout}) = \frac{a-b}{a+b}, \qquad \mathbb{P}(\text{A never behind}) = \frac{a-b+1}{a+1},$$

$$\mathbb{P}(\text{some tie after the 1st vote}) = \frac{2b}{a+b}.$$

Counts: strictly ahead $= \frac{a-b}{a+b}\binom{a+b}{a}$; never behind $= \binom{a+b}{b} - \binom{a+b}{b-1}$.

Paths with a floor. Number of n-step paths from 0 to $e \geq 0$ that never go below 0:

$$\binom{n}{\frac{n+e}{2}} - \binom{n}{\frac{n+e}{2}+1}. \qquad \text{Strictly positive after time 0, ending at } e > 0: \quad \frac{e}{n}\binom{n}{\frac{n+e}{2}}.$$

4.2 Catalan numbers

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1}, \qquad C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}.$$

C_n counts: Dyck paths of length $2n$ (never below 0, end at 0); balanced parentheses with n pairs; non-crossing chord matchings; binary trees; triangulations of an $(n+2)$-gon. Paths of length $2n$ that stay *strictly* positive between 0 and $2n$: C_{n-1}.

$$\mathbb{P}(\text{random } 2n\text{-step walk is a Dyck path}) = \frac{C_n}{4^n}.$$

5. Gambler's ruin: the core formulas

5.1 Equations (first-step analysis)

$$W_k = pW_{k+1} + qW_{k-1}, \quad W_0 = 0,\ W_N = 1; \qquad D_k = 1 + pD_{k+1} + qD_{k-1}, \quad D_0 = D_N = 0.$$

General solution of the first: $W_k = A + B\,r^k$ (biased) or $A + Bk$ (fair).

5.2 Win and ruin probabilities

$$\textbf{Fair: } W_k = \frac{k}{N}, \quad Q_k = \frac{N-k}{N}. \qquad \textbf{Biased: } W_k = \frac{1-r^k}{1-r^N}, \quad Q_k = \frac{r^k - r^N}{1-r^N}, \quad r = \frac{q}{p}.$$

Halfway trick ($N = 2k$): $W_k = \dfrac{1}{1+r^k}$. **Mirror:** replace $(p,k) \leftrightarrow (q, N-k)$ to swap the roles of the two players.

Two-player form: players with stacks m and n ($N = m+n$), M wins each round w.p. p: $\mathbb{P}(M \text{ wins all}) = \dfrac{1-(q/p)^m}{1-(q/p)^{m+n}}$ (fair: $\frac{m}{m+n}$).

Walk form: from 0, fair: $\mathbb{P}(\text{hit } +a \text{ before } -b) = \dfrac{b}{a+b}$. Biased: shift to $k = b$, $N = a+b$ and use the biased formula.

5.3 Expected duration

$$\textbf{Fair: } D_k = k(N-k). \qquad \textbf{Biased (Wald): } D_k = \frac{N\,W_k - k}{p-q} = \frac{k}{q-p} - \frac{N}{q-p}\cdot\frac{1-r^k}{1-r^N}.$$

Fair, exit from $(-a,b)$ starting at 0: $\mathbb{E}[T] = ab$. From 0 to $|S_n| = a$: $\mathbb{E}[T] = a^2$.

Variance of the duration (fair):

$$\mathrm{Var}(T) = \frac{k(N-k)\left[k^2 + (N-k)^2 - 2\right]}{3}.$$

Sanity checks: $\sum$ exit-probability $\times$ conditional time $= D_k$; fair game: $\mathbb{E}[S_T] = k$; biased: $\mathbb{E}[S_T] = k + (p-q)D_k$.

6. One-sided barriers: infinite bank, drift, maximum

6.1 Let $N \to \infty$ (cliff-hanger family)

$$\mathbb{P}(\text{ever hit } 0 \mid \text{start } k) = \min\{1, (q/p)^k\}.$$

Derivation in one line: $P_1 = q + pP_1^2 \Rightarrow (P_1 - 1)(pP_1 - q) = 0$, and one picks $P_1 = \min(1, q/p)$ (the root that stays in $[0,1]$ and is continuous in p).

- $p > \frac{1}{2}$ (up-drift): ruin from k has probability $(q/p)^k$; time to climb a levels: $\mathbb{E}[T_a] = \dfrac{a}{p-q}$, $\mathrm{Var}(T_a) = \dfrac{a \cdot 4pq}{(p-q)^3}$.

- $p < \frac{1}{2}$ (down-drift): ruin is certain, $\mathbb{E}[\text{time to ruin from } k] = \dfrac{k}{q-p}$; the walk reaches $+a$ ever with probability $(p/q)^a$.
- $p = \frac{1}{2}$: ruin is certain but $\mathbb{E}[T] = \infty$ (the fair-walk paradox).

6.2 All-time maximum of a down-drifting walk ($p < q$, $\rho = p/q$)

$$\mathbb{P}(M \geq a) = \rho^a, \qquad \mathbb{P}(M = a) = (1-\rho)\rho^a,$$
$$\mathbb{E}[M] = \frac{\rho}{1-\rho} = \frac{p}{q-p}, \qquad \mathrm{Var}(M) = \frac{\rho}{(1-\rho)^2} = \frac{pq}{(q-p)^2}.$$

Limit-order interpretation: a limit sell at level $+a$ eventually fills with probability ρ^a.

7. Martingale toolkit (optional stopping)

7.1 The four martingales to know

Martingale	Valid when	What it gives
S_n	fair walk	$W_k = k/N$
$S_n - \mu n$	any p (Wald)	$\mathbb{E}[S_T] = k + \mu\,\mathbb{E}[T]$
$S_n^2 - n$	fair walk	$\mathbb{E}[T] = k(N-k)$
r^{S_n}, $r = q/p$	any p	$W_k = \dfrac{1-r^k}{1-r^N}$
$(S_n - \mu n)^2 - \sigma^2 n$	any p	variance of T

Recipe: (1) find a martingale Z_n; (2) check the stopped process is bounded; (3) write $\mathbb{E}[Z_T] = Z_0$ using the two exit values; (4) solve for the unknown.

8. Conditioning, visit counts and excursions

8.1 Conditioning a fair walk on its exit

Conditioned on ending at N, the walk steps up from k with probability $\dfrac{k+1}{2k}$ (a Doob transform), and

$$\mathbb{E}[T \mid \text{win}] = \frac{N^2-k^2}{3}, \qquad \mathbb{E}[T \mid \text{ruin}] = \frac{N^2-(N-k)^2}{3} = \frac{k(2N-k)}{3}, \qquad \mathbb{E}[T \cdot \mathbf{1}_{\text{win}}] = \frac{k(N^2-k^2)}{3N}.$$

Check: $\frac{k}{N} \cdot \frac{N^2-k^2}{3} + \frac{N-k}{N} \cdot \frac{k(2N-k)}{3} = k(N-k)$.

8.2 Expected visits and the escape-probability trick

Geometric trick: $\mathbb{E}[\#\text{visits to a state}] = \dfrac{1}{\mathbb{P}(\text{escape before returning})}$.

Fair walk, absorbing at 0 and N: expected visits to j starting from k:

$$G(k,j) = \frac{2\min(k,j)\,\big(N - \max(k,j)\big)}{N}.$$

Excursions from 0 (fair):

$$\mathbb{P}(\text{reach } +a \text{ before returning to } 0) = \frac{1}{2a}, \qquad \mathbb{P}(\text{reach } \pm a \text{ before returning}) = \frac{1}{a},$$
$$\mathbb{E}[\#\text{visits to } 0 \text{ before } |S| \text{ hits } a] = a.$$

9. Variants: ties, bet sizes, walls and cycles

9.1 Reductions (turn a weird game into the standard one)

- **Ties are pauses.** Win p, lose q, tie $t = 1-p-q$. Use the embedded walk with $p' = \frac{p}{p+q}$, $q' = \frac{q}{p+q}$; the win probability is unchanged and the duration is multiplied by $\dfrac{1}{p+q}$. Fair with ties: $D_k = \dfrac{k(N-k)}{\sigma^2}$, $\sigma^2 = p+q$.
- **Bet size b.** Replace $k \to k/b$, $N \to N/b$ (stakes in units of the bet). Against an unfavourable game ($p < \frac{1}{2}$) a bigger bet means a higher chance to reach the target: bold play is best.
- **Bold vs timid, target $2\times$ start:** bold (one bet) wins w.p. p; timid (unit bets, k units) wins w.p. $\dfrac{1}{1+r^k}$.
- **Different goals/stops:** P&L barriers at $+A$ and $-B$ from 0: set $k = B$, $N = A+B$.

9.2 Reflecting barrier at 0, absorbing at N

Unfolding: $|S_n|$ is a walk reflecting at 0. For the *fair* walk with a hard wall ($0 \to 1$ w.p. 1):

$$D_k = N^2 - k^2 \qquad (\text{from } 0:\ N^2).$$

Sticky wall (stay at 0 w.p. $\frac{1}{2}$): $D_k = (N-k)(N+k+1)$, from 0: $N(N+1)$.
Biased wall (hard wall at 0, $\mu = p-q \neq 0$):

$$D_k = \frac{N-k}{p-q} - \frac{2pq}{(p-q)^2}\left(r^k - r^N\right).$$

9.3 Cycle $\mathbb{Z}_N$ (fair steps)

$$\mathbb{E}[\text{time to hit a vertex at distance } d] = d(N-d), \qquad \mathbb{E}[\text{cover time}] = \frac{N(N-1)}{2},$$

$$\mathbb{P}(\text{vertex } j \text{ is the last visited}) = \frac{1}{N-1} \quad (j \neq \text{start}).$$

The cover time comes from $\sum_{j=1}^{N-1} j$: with j vertices visited the ant needs j steps on average to reach a new one. Expected return time to the start is N.

10. Higher dimensions

10.1 Return probabilities on $\mathbb{Z}^d$ (simple walk, $2d$ neighbours)

d	$\mathbb{P}(\text{origin at time } 2n)$	After 4 steps	$\mathbb{P}(\text{ever return})$
1	u_{2n}	$\frac{3}{8}$	1
2	u_{2n}^2	$\frac{9}{64}$	1
3	$\binom{2n}{n}4^{-n}\sum_{i+j+k=n}\left(\frac{n!}{i!j!k!3^n}\right)^2$	$\frac{5}{72}$	≈ 0.3405

Large n: $\mathbb{P}(\text{origin at } 2n) \approx 2\left(\dfrac{d}{4\pi n}\right)^{d/2}$, so the series $\sum_n$ diverges for $d = 1, 2$ and converges for $d \geq 3$.
Recurrence criterion: $\sum_n \mathbb{P}(\text{at origin at } n) = \infty \iff$ return probability $= 1$. If the sum is finite, $P_{\text{return}} = 1 - \dfrac{1}{\sum_{n\geq 0}\mathbb{P}(\text{at origin at } n)}$, equivalently $\mathbb{E}[\#\text{visits}] = \dfrac{1}{1-P_{\text{return}}}$.
Closed 4-step walks: $\mathbb{Z}$: 6; $\mathbb{Z}^2$: $\binom{4}{2}^2 = 36$; $\mathbb{Z}^3$: 90. (Total walks: 2^4, 4^4, 6^4.)
Diagonal 2D / cube-corner 3D walks (each coordinate an independent fair ± 1 step): $\mathbb{P}(\text{origin at } 2n) =$

u_{2n}^2 in 2D and u_{2n}^3 in 3D. The 3D cube-corner walk returns with probability ≈ 0.282.

11. Number tables

Central binomials, Catalan numbers and u_{2n}

n	0	1	2	3	4	5	6	7	8	9	10
$\binom{2n}{n}$	1	2	6	20	70	252	924	3432	12870	48620	184756
C_n	1	1	2	5	14	42	132	429	1430	4862	16796
u_{2n}	1	.500	.375	.313	.273	.246	.226	.209	.196	.185	.176

Handy values: $\ln(10/9) = 0.10536$, $\ln 2 = 0.6931$, $\ln(9/11) = -0.2007$, $(2/3)^5 = 32/243$, $\binom{20}{10} = 184756$, $2^{10} = 1024$, $2^{20} = 1048576$.

12. Which formula do I use? (decision table)

If the question says...	Use...
Fair game, two absorbing ends, "probability of winning"	$W_k = k/N$ (stack ÷ total money)
Fair game, "expected number of rounds/ticks"	$k(N-k)$; with ties or lazy steps divide by σ^2
Biased coin, two ends	$\dfrac{1-r^k}{1-r^N}$; if the start is exactly halfway $\dfrac{1}{1+r^k}$
Biased, duration	Wald: $\dfrac{NW_k - k}{p-q}$
Unlimited bank / "ever reaches" / "cliff"	$\min\{1, (q/p)^k\}$; geometric ladder of levels
Expected time to climb to $+a$ with up-drift	$a/(p-q)$
"Maximum ever" of a down-drifting walk	geometric: $\mathbb{E}[M] = \frac{p}{q-p}$
"Touches level m in n steps"	reflection: $2\mathbb{P}(S_n > m) + \mathbb{P}(S_n = m)$
"Touches m and ends below c"	count the mirror endpoints $2m - e$
"Never goes negative" / "never behind"	ballot/Catalan: $\binom{n}{j} - \binom{n}{j-1}$ or $\frac{a-b+1}{a+1}$
"A strictly ahead" with random order	$\frac{a-b}{a+b}$
"Tied at some time"	$\frac{2b}{a+b}$
"First return to 0 at time $2n$"	$u_{2n}/(2n-1) = 2C_{n-1}/4^n$
"No return by time $2n$"	$u_{2n} = \binom{2n}{n}4^{-n}$
Reflecting wall at 0	unfold with $\lvert S_n\rvert$: $N^2 - k^2$ (fair)
Walk on a cycle / clock / polygon	$d(N-d)$, cover $\frac{N(N-1)}{2}$, last vertex uniform
Expected visits to a state	(probability to reach it) $\times$ 1/(escape probability), or $G(k,j)$
"Given that he wins... how long?"	$(N^2-k^2)/3$ (conditioned walk)
2D / 3D return	u_{2n}^2 (2D), recurrence iff $\sum$ diverges
Ties in the game	delete the ties: embedded walk, rescale time by $1/(p+q)$

13. Which formula solves which of your problems

Map to the 28 study-note examples (file: `random_walks_gamblers_ruin_notes.tex`)

Formula	Examples
Fair ruin k/N, duration $k(N-k)$ (5.2, 5.3)	1, 2
Biased ruin, halfway trick (5.2)	3, 4, 5, 6
Biased duration, Wald (5.3)	4, 9
Ties are pauses, lazy walk (9.1)	7
Martingale derivation (7.1)	8
Infinite bank, cliff, maximum, drift time (6.1, 6.2)	10, 11, 12, 13
Biased return probability (2.2)	14
Reflecting wall, unfolding (9.2)	15
Cycle formulas (9.3)	16
Expected visits, excursions (8.2)	17, 18
Path counting (1.1)	19
Reflection principle (3.1)	20
Catalan numbers (4.2)	21
Ballot theorem (4.1)	22, 23
Returns / no-tie probabilities (2.1)	24, 25, 26
Higher dimensions (10.1)	27, 28

Map to the 10 hard Stage-1 problems (file: `random_walks_10_hard_stage1_problems.tex`)

Formula	Problems
Fair ruin, duration, martingale EV (5.2, 5.3, 7.1)	1
Biased ruin, halfway trick, Wald (5.2, 5.3)	2, 9
Conditioned exit time (8.1)	3
Infinite bank, all-time maximum (6.1, 6.2)	4
Reflection principle with endpoints (3.1)	5
Ballot, Catalan, first return (2.1, 4.1, 4.2)	6
Reflecting and sticky wall (9.2)	7
Cycle formulas, uniform last vertex (9.3)	8
Dimensions, recurrence (10.1)	10

14. Last-minute error list

Top mistakes to avoid

1. **Mixing up $r = q/p$ and p/q.** In $\frac{1-r^k}{1-r^N}$ the ratio is (lose)/(win). If $p > \frac{1}{2}$, then $r < 1$ and $W_k \to 1 - r^k$ as $N \to \infty$.
2. **Parity.** A walk of n steps only reaches levels with the same parity as n.
3. **Wrong shift.** For barriers $+A$ and $-B$ from 0, use $k = B$ and $N = A + B$ (not the other way round).
4. **"Never behind" vs "strictly ahead".** $\frac{a-b+1}{a+1} \neq \frac{a-b}{a+b}$.
5. **Forgetting to double** after the reflection principle: it is $2\mathbb{P}(S_n > m) + \mathbb{P}(S_n = m)$, not $\mathbb{P}(S_n \geq m)$.
6. **Treating a conditioned walk as fair.** $\mathbb{E}[T \mid \text{win}] \neq \mathbb{E}[T]$.
7. **Splitting bets does not help** when $p < \frac{1}{2}$: smaller bets reduce, not raise, the chance of reaching a

target.

8. **Forgetting ties** change the duration but not the win probability.